

Example Final Exam Problems

This is just to give you some ideas about the type of problems that will appear in the final exam.

None of the problems requires long answers; a couple clearly stated sentences are enough.

When you review the lecture slides, do not worry about memorize the equations. I will list the equations when asking about them in the exam; your job is to understand the meanings and purposes of the different items in the equations.

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Pretext tasks are tasks that are used for representation learning with self-supervision. Give one example each of pretext tasks for (1) image data, and (2) natural language data.

What is the difference between model-based and model-free reinforcement learning methods?

In reinforcement learning, use your own words to explain what a "policy" is. Also explain why the state values dependent upon the policy?

What is the bias-variance dilemma (or trade-off), and explain how this is related to model complexity in supervised learning.

Sketch an autoencoder. Also point out the "information bottleneck" and briefly describe its purpose.

Given the following cost function of support vector machines, explain the meanings and/or purposes of the two terms.

$$J = \frac{1}{2} \| \mathbf{w} \|^2 + C \sum_{i=1}^N \xi_i$$

In principle component analysis (PCA), the sample vectors are projected to a space of lower dimensionality. Explain the criteria used to determine this projection in PCA, i.e., what is minimized or maximized.

Use the figure to the right to describe the basic idea of contrastive learning.

